Engineering Chemistry (AHT-002)

Complete unit-wise notes - easy English, matched to the UTU syllabus.

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Unit 1 - Atomic & Molecular Structure

What this unit covers (syllabus): atomic theory and nuclear models, Heisenberg's uncertainty principle, de Broglie waves, the Schrödinger equation for hydrogen, VBT and MOT of diatomic molecules (H_2, O_2, N_2, H_2O, NH_3, XeO_3, XeO_4), crystal field theory and magnetic properties of transition metal ions, and band theory of metals.

Key ideas: de Broglie: every moving particle has a wave, $\lambda = h/mv$. **Heisenberg:** $\Delta x \cdot \Delta p \geq h/4\pi$ - position and momentum cannot both be exact. The **Schrödinger equation** ($H\psi = E\psi$) gives orbitals as solutions. **VBT**: bonds form by orbital overlap, explained with hybridisation (sp^3 in NH_3/H_2O). **MOT:** atomic orbitals combine into bonding (lower energy) and antibonding (higher) molecular orbitals; **bond order** = (bonding - antibonding electrons)/2 - O_2 has bond order 2 and is paramagnetic (two unpaired electrons), the classic question. **Crystal field theory:** d-orbitals split in a ligand field (t_2g/e_g in octahedral); strong-field ligands pair electrons (low spin), weak-field leave them unpaired (high spin) - this decides magnetism. **Band theory:** metals conduct because their valence and conduction bands overlap.

Exam pointers: MO diagram of O_2 or N_2 with bond order and magnetism is the most repeated question. de Broglie/Heisenberg numericals (plug into the formula) and "VBT vs MOT" are regular theory.

Unit 2 - Thermodynamics & Electrochemistry

What this unit covers (syllabus): first and second laws of thermodynamics; energy, entropy and free energy; EMF and cell potential; the Nernst equation and applications; acids and bases; Ellingham diagrams in metallurgy; Hess's law.

Key ideas: First law: energy is conserved - $\Delta U = q + w$. **Second law:** entropy of the universe increases for a spontaneous process. **Gibbs free energy** $\Delta G = \Delta H - T\Delta S$ decides spontaneity: $\Delta G < 0$ spontaneous. **Electrochemistry:** a galvanic cell converts chemical energy to electricity; **EMF** = E(cathode) - E(anode), and $\Delta G = -nFE$ connects the two halves of the unit. The **Nernst equation** $E = E° - (0.0591/n) \log Q$ (at 25°C) corrects for concentration - the standard numerical. **Hess's law:** total enthalpy change is the same whatever the path - add/subtract given equations to reach the target. **Ellingham diagram** plots $\Delta G°$ vs T for oxide formation; the line lower at a given temperature reduces the oxide above it - this explains choice of reducing agents in metallurgy.

Exam pointers: one Nernst-equation numerical (calculate EMF at given concentrations), $\Delta G = \Delta H - T\Delta S$ spontaneity checks, and a Hess's law calculation are the regulars. "Explain the Ellingham diagram and its use in metallurgy" is a frequent long answer.

Unit 3 - Water Chemistry & Corrosion

What this unit covers (syllabus): hardness of water and its treatment (Clark's, lime-soda, zeolite, reverse osmosis), industrial water treatment, alkalinity, and corrosion - theories, types, mechanism and control.

Key ideas: Hardness: temporary (bicarbonates of Ca/Mg - removed by boiling or Clark's method with lime) and permanent (chlorides/sulphates - need lime-soda, zeolite or ion exchange). The **lime-soda process** adds $Ca(OH)_2$ and Na_2CO_3 to precipitate $CaCO_3$ and $Mg(OH)_2$; the **zeolite process** swaps Ca^{2+}/Mg^{2+} for Na^+ (regenerated with brine); **reverse osmosis** pushes water through a membrane against osmotic pressure. Hardness is expressed as $CaCO_3$ equivalents: mg/L = (mass of salt x 100 x 1000)/(MW of salt x volume). **Corrosion** = metal eaten by its environment: the **electrochemical theory** sets up tiny anode-cathode cells on the metal surface (rusting: Fe oxidises at the anode, O_2 reduces at the cathode). Types: galvanic, pitting, stress, water-line. Control: coatings, galvanisation (sacrificial zinc), cathodic protection, inhibitors.

Exam pointers: hardness numericals (convert to $CaCO_3$ equivalents, or lime/soda requirement) appear almost every year - practise the formula. "Explain the electrochemical theory of rusting with the anode/cathode reactions" and "compare lime-soda and zeolite" are the standard theory answers.

Unit 4 - Materials: Polymers, Fuels, Lubricants, Glass

What this unit covers (syllabus): materials - polymers, fuels, lubricants and glass/cement basics.

Key ideas: Polymers: addition (chain-growth - polyethylene, PVC) vs condensation (step-growth - nylon, bakelite, with a small molecule eliminated). Thermoplastics soften on heating (recyclable); thermosets set permanently. **Fuels:** calorific value (higher and lower), characteristics of a good fuel, knocking and **octane number** for petrol, **cetane number** for diesel; gaseous fuels (CNG, LPG, biogas) compared on calorific value. **Lubricants:** reduce friction and wear; classified solid/semi-solid/liquid with the key properties (viscosity, viscosity index, flash point) - full details in the notes written under this unit. **Glass:** a supercooled liquid of silicates; types (soda-lime, borosilicate) and annealing.

Exam pointers: "addition vs condensation polymerisation with two examples each" and "thermoplastic vs

thermoset" tables are free marks. Octane vs cetane numbers and "properties of a good lubricant" are the short-answer regulars.

Extra note - Lubricants - quick exam notes:

Lubricant: a substance between two moving surfaces that reduces friction and wear.

Functions: reduces friction and wear, carries away heat, seals out dust, prevents corrosion, and cushions shock between parts.

Classification: (1) **Solid** - graphite, molybdenum disulphide (used where oils would catch fire or leak); (2) **Semi-solid** - greases (oil + soap thickener; for slow, heavy loads); (3) **Liquid** - mineral oils (petroleum), vegetable and animal oils, blended oils (mineral oil + additives for viscosity index, oxidation stability).

Key properties: viscosity and viscosity index (how much viscosity changes with temperature), flash and fire point, cloud and pour point (low-temperature flow), oiliness (sticking to the surface), acid value, saponification value.

Exam tip: definitions + one example each of solid/semi-solid/liquid lubricant, and the meaning of viscosity index - this pattern repeats in papers.

Unit 5 - Spectroscopy, Organic Reactions & Drug Synthesis

What this unit covers (syllabus): spectroscopy (principle, selection rules, electronic/vibrational/rotational), NMR and MRI, organic reaction types (addition, elimination, substitution, cyclization - Diels-Alder), and synthesis of one drug (aspirin, phenacetin, melubrin, novalgin).

Key ideas: Spectroscopy studies how molecules absorb electromagnetic radiation: rotational (microwave), vibrational (infrared), electronic (UV-visible) - energy gaps decide the region. **Selection rules** say which transitions are allowed (vibrational needs a changing dipole moment; rotational needs a permanent dipole). IR is used to identify functional groups from characteristic stretching frequencies. **NMR:** nuclei with spin (^{1}H) flip between spin states in a magnetic field; chemical shift tells the proton's environment, splitting (n+1 rule) tells the neighbours. **MRI** applies the same physics to image body tissue - see the simple notes under this unit. **Organic reactions:** addition (across double bonds), elimination (form double bonds), substitution (swap a group), and the **Diels-Alder** cycloaddition (diene + dienophile → six-membered ring). **Aspirin synthesis:** salicylic acid + acetic anhydride (esterification of the -OH group).

Exam pointers: "principle of NMR spectroscopy" and "application of IR" are the common theory questions; one Diels-Alder mechanism and the aspirin synthesis (equation with conditions) appear regularly. Draw the MO-free, arrow-pushing steps cleanly - mechanism marks are step marks.

Extra note - MRI - made simple:

MRI (Magnetic Resonance Imaging) = a machine that photographs the inside of the body, working on the principle of **NMR spectroscopy**.

> *Just as NMR finds the structure of small molecules by applying a magnetic field, MRI maps the whole body (mostly the hydrogen atoms in water).*

- Hydrogen (in water) is everywhere in the body. A strong magnetic field + radio waves are applied.
- Hydrogen in each tissue gives a different signal - a computer turns that into an image.
- **Advantage:** no harmful radiation (unlike X-rays), and a clear image of soft tissue.

Exam tip: "MRI is an application of NMR" - do not forget to write this one line.